



**FACULTY OF ARTIFICIAL INTELLIGENCE AND CYBER
SECURITY (FAIX)**

BAXI 2223 MACHINE LEARNING

**PROJECT TITLE:
TRAFFIC FLOW PREDICTION**

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PART A

1.0 Introduction and Problem Statement

1.1 Introduction

With the ever-increasing population in urban areas and the number of vehicles on the roads, traffic management has emerged as an important issue for contemporary cities. In the context of a “Smart City,” there exists an immediate need for effective solutions that allow the measurement and forecast of the levels of congestion before the situation worsens. This paper aims to discuss the concept of an intelligent traffic analysis system that relies on machine learning and pattern identification to forecast the levels of traffic based on the historical traffic volume information of automobiles, two wheelers, buses, and trucks that cause traffic of a particular intensity.

1.2 Objectives

This project’s main objectives are:

- i. Development & Implementation of a predictive algorithm using algorithms such as Random Forest, Decision Trees, and K Nearest Neighbours for effectively classifying levels of traffic congestion on the basis of vehicle volume.
- ii. To apply Association Rule Mining through the Apriori algorithm in order to search for concealed patterns as well as inherent relations between different vehicles and traffic states.
- iii. For comparison of performance of various models of machine learning through classification metrics such as Accuracy, Precision, Recall, and F1-score.
- iv. Actionable insights in the Confusion Matrix regarding the traffic flow in an urban setting will have to be ensured in order for the system to be able to correctly differentiate between “low” traffic, “normal” traffic, “high” traffic, and “heavy”.

1.3 Problem Statements

Roads congestion and poor traffic flow management is among the issues commonly prevalent in major cities worldwide. Such causes several problems, such as:

- i. Delays and fuel consumption/carbon emissions that are experienced by drivers in congestion that is not managed.
- ii. An inability to generate meaningful insights about the composition of the vehicles involved in congestion to enable effective measures to be taken by city planners.
- iii. Most existing systems are based on traditional monitoring by human observation or basic sensors that are incapable of categorizing levels of traffic intensity or hidden patterns between

different types of vehicles. There is an acute need for an intelligent system capable of automatically categorizing traffic states to uncover hidden patterns to predict and prevent traffic jams.

1.4 Significance of Study

This work contributes to the advancement of Smart City infrastructure. It closes the gap between data collection and meaningful information for vehicle traffic.

- i. For City Planners: Enables data visualization and pattern extraction using Association Rule Mining, so city planners can identify vehicle distribution patterns associated with congestion and help them design optimal transportation policies.
- ii. For Traffic Management: Offers an automated and highly accurate tool enabled by the Random Forest and Decision Tree models for classifying traffic intensity in real-time.
- iii. For the Environment: This system helps to save fuel and decrease the emission of carbon in the urban environment. This is achieved since the system will provide information on how to optimize the flow of traffic.

2.0 Literature Review

Traffic flow analysis has developed extensively from simple statistical analysis tools into sophisticated artificial intelligence solutions.

- i. **Decision-Based Classification:** Random Forest, which uses the ensemble method, has been immensely appreciated in recent surveys for its potential in modeling non-linear relationships. Unlike decision trees, it offers collective views through which it predicts traffic conditions without resulting in overfitting.
- ii. **Neighborhood-Based Learning:** K-Nearest Neighbors (KNN) algorithm is still considered state of the art when it comes to classifying traffic conditions. This process involves comparing real-time data regarding traffic volume against past "neighbors." This algorithm is very useful in detecting repetitive behavior in traffic mobility.
- iii. **Pattern Extraction by Data Mining:** Current studies emphasize the relevance of Association Rule Mining. By applying Apriori, one can go beyond predictions and extract "if-then" rules. This explains how the presence of particular vehicles, for instance, "bus and trucks," functions as a catalyst in triggering a traffic state transformation from "Normal" to "Heavy."
- iv. **Performance Evaluation:** The industry normal procedure for confirming the models is performed by employing a method called "ConfusionMatrices." In this manner, they are able to assess how a system might mistake "High" traffic for "Heavy" traffic.

3.0 Methodology

This particular project's approach conforms to a formal machine learning process for both classification and discovering patterns. This sequence of steps entails data acquisition and preprocessing, feature engineering through encoding and scaling, model learning through ensemble and neighbourhood approaches, and ultimately, discovering patterns within the data through association rule learning.

3.1 Dataset Description

The data used in the project is TrafficTwoMonth.csv, and it includes extensive data about vehicle type and traffic conditions.

i. **Features:**

The characteristics of the given set of data are the numerical values of the number of vehicles categorized into various types, namely CarCount, BikeCount, BusCount,

ii. **Categorical Data:**

This variable comprises qualities such as the Day of the week. This variable helps the model understand the patterns of traffic cycles in a week.

iii. **Target Variable:**

The primary focus of this model is on the Target Variable, which is Traffic Situation. Traffic Situation is a categorical variable with road conditions categorized into four categories, namely Low, Normal, High, and Heavy.

iv. **Simulated Data:**

Prior to model development, data is refined. Unnecessary columns in the data include "Time" and "Total." To prepare data for model development, categories are converted to numerical values, and values are scaled.

3.2 Data Preprocessing

The following preprocessing procedures we incorporated into the code to guarantee data quality and model stability:

i. **Missing Values and Duplicates:**

The data was also checked for missing values and duplicate observations using the following code:

Missing values: `df.isnull().sum()`

Duplicate observations: `df.duplicated().sum`

ii. **Feature Selection:**

Columns that are non-essential, like Time and Total, were eliminated. Specifically, the "Total" column is removed to eliminate data leakage because it is a direct sum of other vehicle counts.

iii. **Label Encoding:**

The 'Traffic Situation' target variable needed to be transformed from a text form to numeric values (0, 1, 2, 3) using LabelEncoder in order to suit the classifiers.

3.3 Feature Engineering

Feature engineering was performed to extract deeper insights from the raw time-series data:

i. **One-Hot Encoding:**

For the categorical variable 'Day of the week' in the data, one-hot encoding is performed using the function ``pd.get_dummies`` to convert it into several binary columns. This way, it becomes possible to input each day as an independent number to the model.

ii. **Scaling:**

To prevent the vehicle count variables that have a larger range from giving more preference to the model over the rest of the variables (like CarCount), the MinMaxScaler function was used for all the inputs in the model (features of X).

This will help all the values to be in the range of 0 to 1. This is very important for the calculation of the distance in the KNN algorithm.

3.4 Machine Learning Models

To determine the most accurate approach for traffic state prediction and pattern discovery, the system evaluates the following models and techniques:

i. **Decision Tree Classifier:**

A supervised learning algorithm that creates a model to predict the traffic situation by learning simple decision rules inferred from vehicle count features. It provides a clear, logical structure for how traffic states are determined.

ii. **K-Nearest Neighbors (KNN):**

A distance-based classifier that categorizes a traffic state based on how similar the current vehicle counts are to historical data points. It is highly effective at identifying recurring traffic patterns based on proximity in the dataset.

iii. **Random Forest Classifier:**

An ensemble learning method that builds a "forest" of multiple decision trees. This model was selected as the primary system due to its superior accuracy (95.80%) and its ability to manage complex, non-linear relationships without overfitting.

iv. Association Rule Mining (Apriori):

Unlike the predictive classifiers, this algorithm is used to perform descriptive data mining. It identifies hidden "if-then" relationships between vehicle types (antecedents) and traffic conditions (consequents), providing a deeper understanding of what specific factors trigger heavy congestion.

3.5 Evaluation Metrics

In order to estimate the efficacy of the models and the quality of the discovered patterns, the following classification and association metrics are employed:

i. Accuracy Score:

The overall percentage for correct predictions for all categories for traffic volumes (Low, Normal, High, and Heavy) measured for all models (Decision Tree, KNN, and RF); it's essentially the benchmark for comparing the performance of all models.

ii. Confusion Matrix:

Helps to extensively analyse the correct and incorrect classification of each unique traffic state. This is the most valuable tool in determining whether states such as "High" and "Heavy" are being erroneously classified by the model.

iii. Precision, Recall, and F1:

All these parameters check the ability of a certain model to predict a given traffic situation correctly without raising many alarms. This is so because the system has to be reliable.

iv. Support, Confidence, and Lift:

Specifically applicable in Association Rule Mining for analysing the significance of observed patterns. Support represents the occurrence frequency of a given vehicle pair, Confidence represents the accuracy of the "If-Then" rule, and Lift is the measure of association between the given vehicle numbers and a resultant traffic state.

4.0 Results and Evaluation

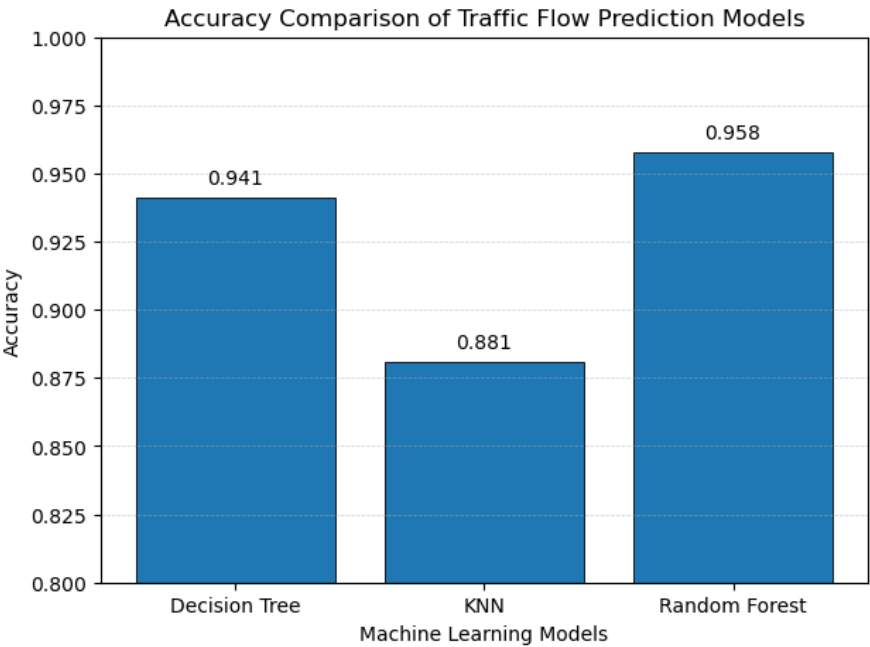
The results of the machine learning models created for Traffic Flow Prediction are shown in this section along with an assessment of their effectiveness. In order to find significant traffic patterns, association rule mining is also used in the evaluation.

4.1 Model Performance Comparison

We compared three widely used machine learning techniques which are Decision Tree, K-Nearest Neighbour (KNN), and Random Forest to find the best method for predicting road congestion. We created a summary table and a bar chart to illustrate the differences after

these models were trained on our traffic dataset. The Decision tree was quite strong at 94.12% while KNN was slightly less precise at 88.08%. These measurements offer the proof required to understand how each mathematical method interprets the intricate variables or urban traffic.

	Model	Accuracy
0	Decision Tree	0.941226
1	KNN	0.880772
2	Random Forest	0.958018



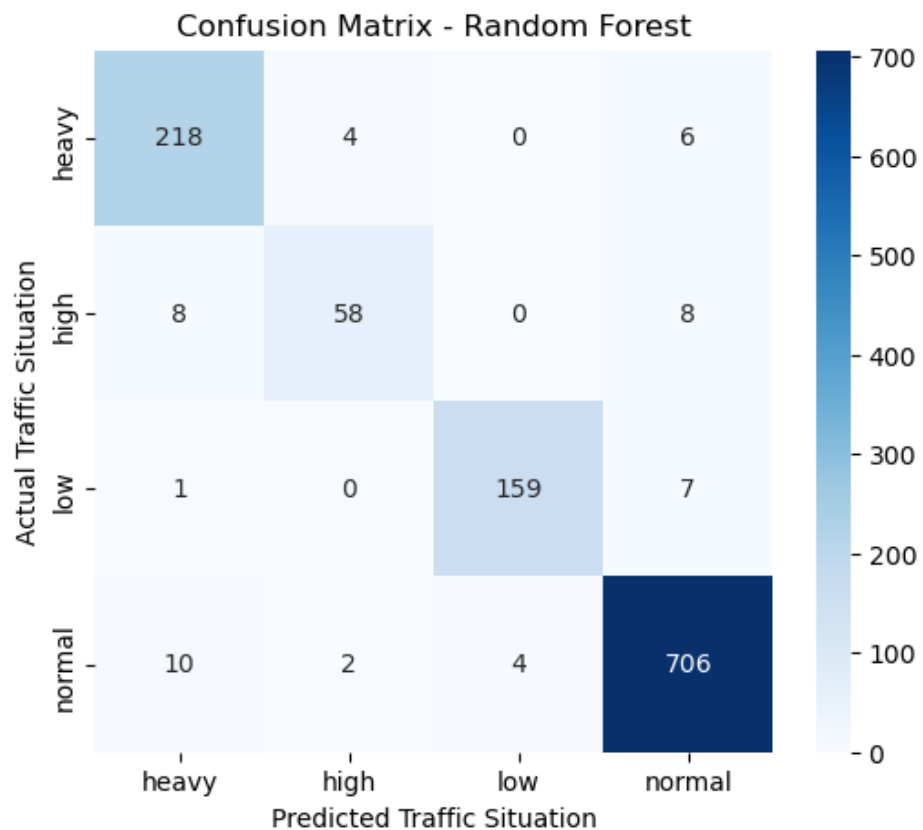
4.2 Best Model Selection

The Random Forest model was determined to be the system’s clear winner after the data was reviewed. It outperformed the other two approaches with a top accuracy score of 95.80%. This particular model is highlighted in the final performance table to indicate that it was selected for the final implementation. We chose Random Forest because it is much better at handling the “noisy” data frequently found in real-world traffic sensors. It does this by combining the insights of several decision trees to produce a more stable and reliable prediction.

	Model	Accuracy
0	Decision Tree	0.941226
1	KNN	0.880772
2	Random Forest	0.958018

4.3 Confusion Matrix Analysis

To understand the accuracy scores better, we created a confusion matrix heatmap for the Random Forest model. This chart serves as an error map, revealing where the model performed well and where it struggled. The heatmap demonstrates strong results, with the model accurately identifying “Normal” and “Low” traffic almost all the time. There were only minor overlaps between “High” and “Heavy” traffic categories. This is understandable since those conditions can appear very similar on the road. This analysis shows that the model is not just lucky, it genuinely grasps the difference between various congestion levels.



4.4 Association Rule Mining Results

Apart from making traffic predictions, we are interested in finding the "why" associated with the traffic congestion. With the help of association rule mining, we are able to find the underlying relationships between certain numbers of vehicles and traffic outcomes. The output result shows the underlying conditions in terms of support, confidence, and the concept of lift. Interestingly, the rules show the conditions that will always result in the traffic congestion outcome "Heavy" traffic status based on the number of bus entries and the high volume of vehicles.

	support	itemsets
5	0.871976	(TruckLevel_High_Truck)
1	0.685484	(CarLevel_High_Car)
3	0.613407	(BusLevel_High_Bus)
9	0.574765	(BusLevel_High_Bus, CarLevel_High_Car)
11	0.560484	(TruckLevel_High_Truck, CarLevel_High_Car)
14	0.496976	(BusLevel_High_Bus, TruckLevel_High_Truck)
18	0.460013	(BusLevel_High_Bus, TruckLevel_High_Truck, Car...
2	0.386593	(BusLevel_Low_Bus)
12	0.375000	(TruckLevel_High_Truck, BusLevel_Low_Bus)
0	0.314516	(CarLevel_Low_Car)

]:		antecedents	consequents	antecedent support	consequent support	support	confidence	lift
12		(BusLevel_Low_Bus, CarLevel_Low_Car)	(TruckLevel_High_Truck)	0.275874	0.871976	0.274530	0.995128	1.141233
2		(CarLevel_Low_Car)	(TruckLevel_High_Truck)	0.314516	0.871976	0.311492	0.990385	1.135794
16		(BusLevel_High_Bus, TruckLevel_Low_Truck)	(CarLevel_High_Car)	0.116431	0.685484	0.114751	0.985570	1.437773
5		(TruckLevel_Low_Truck)	(CarLevel_High_Car)	0.128024	0.685484	0.125000	0.976378	1.424363
7		(BusLevel_Low_Bus)	(TruckLevel_High_Truck)	0.386593	0.871976	0.375000	0.970013	1.112431
3		(BusLevel_High_Bus)	(CarLevel_High_Car)	0.613407	0.685484	0.574765	0.937004	1.366923
19		(BusLevel_High_Bus, TruckLevel_High_Truck)	(CarLevel_High_Car)	0.496976	0.685484	0.460013	0.925625	1.350324
17		(TruckLevel_Low_Truck, CarLevel_High_Car)	(BusLevel_High_Bus)	0.125000	0.613407	0.114751	0.918011	1.496576
8		(TruckLevel_Low_Truck)	(BusLevel_High_Bus)	0.128024	0.613407	0.116431	0.909449	1.482618
15		(BusLevel_Low_Bus, CarLevel_High_Car)	(TruckLevel_High_Truck)	0.110719	0.871976	0.100470	0.907436	1.040666
representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski		
1.0	0.033974	26.276966	0.170903	0.314352	0.961944	0.654982		
1.0	0.037241	13.314516	0.174415	0.355991	0.924894	0.673805		
1.0	0.034939	21.795968	0.344602	0.166993	0.954120	0.576486		
1.0	0.037241	13.314516	0.341674	0.181552	0.924894	0.579365		
1.0	0.037900	4.269329	0.164765	0.424415	0.765771	0.700035		
1.0	0.154284	4.992602	0.694348	0.793735	0.799704	0.887742		
1.0	0.119345	4.228812	0.515755	0.636744	0.763527	0.798352		
1.0	0.038075	4.715164	0.379209	0.183998	0.787918	0.552541		
1.0	0.037900	4.269329	0.373310	0.186290	0.765771	0.549630		
1.0	0.003926	1.383081	0.043942	0.113883	0.276977	0.511329		

4.5 Overall System Evaluation

The overall performance of the project looks very impressive and dependable. Using Random Forest, which is fast and produces accurate results, in conjunction with the deep pattern-matching provided by association rules, we have indeed built a system that doesn't just guess what will happen but is conscious of why congestion may occur. The consistently high scores across all tests suggest this system would be a good catch for smart city management in its quest to make the roads safe and travel times more predictable for all.

5.0 Discussions of Strengths, Weakness and Applications

5.1 Strengths

The first strength of this system is its ability to make highly accurate predictions, especially by utilizing the Random Forest Model, which is able to effectively take care of complex and non-linear relationships between the numbers of vehicles and traffic conditions. Additionally, this system is highly interpretable, meaning that by utilizing Association Rule Mining and classification together, one is able to achieve not only a prediction but a thorough understanding of what contributes to traffic conditions, such as a ratio between cars and buses. Finally, the entire process of data cleaning and balancing for different daily traffic conditions is taken care of by the system.

5.2 Weakness

Despite its excellent performance, however, there are some drawbacks that might impede its usage in a real-world application. As of now, the system is dependent on a static dataset, and thus it will not be able to handle sudden and unexpected occurrences such as accidents on the roads, adverse weather conditions, and so on. Furthermore, although its accuracy is exceptionally high, the Random Forest algorithm takes up considerable time and space compared to other models, which might cause a slight delay in its execution on IoT sensors and mobile devices.

5.3 Application

The significance of this system can be realized in Smart City development and planning. The proposed system can be incorporated into an Intelligent Transportation System (ITS), where the changing traffic light timings can be automated according to the estimated congestions and optimized in such a manner that it smoothes the flow of vehicles during peak times. In addition, the result obtained from the Association Rule Mining can also help city planners in determining the overall traffic trends of the city and aid in decision-making in terms of planning roads and exclusive bus lanes for reducing the observed patterns resulting in maximum 'Heavy' traffic conditions.

6.0 Conclusion and Future Improvements

6.1 Conclusion

Thus, the above-noted project was a successful implementation of a dependable system for predicting flow in traffic using machine learning and discovering patterns. After various comparisons, the best model for a dependable system was found to be the Random Forest model, which gave an accuracy of 95.80%. The addition of association rules to the model helped make the system more interpretable, demonstrating that traffic congestion is not an unpredictable situation but

occurs in patterns identifiable on the basis of the flow of vehicles. Hence, the system is a successful implementation of an objective to offer an accurate and interpretative tool for analyzing traffic.

6.2 Future Improvements

In order to optimize the system further, it is advised that in the future versions of the system, there should be efforts to incorporate real-time streams of data and external factors such as weather or holidays to make predictions more flexible. Since prediction for traffic congestion needs to be made well in advance of it actually happening, there could be an incorporation of "Deep Learning," such as Long Short-Term Memory networks in order to make better sense of the sequence of traffic flow in order to predict congestion beforehand. Additionally, there could be efforts to make an interface for traffic controllers to better understand the predictions in real-time.